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Quality Management Framework: VI Business (India)

1. Executive Purpose and Document Scope

The Quality Management Framework (QMF) formally establishes the rigid testing methodologies, peer-review checkpoints, and audit standards enforced by VI Business (India) across the entire service delivery lifecycle.

This document assures enterprise customers that large-scale deployments (e.g., rolling out 5,000 SD-WAN edge nodes) are governed by systematic, ISO 9001:2015 certified quality controls, entirely eliminating ad-hoc engineering errors and guaranteeing "First-Time-Right" (FTR) delivery.

2. Definitions, Assumptions, and Applicability

Definitions:

- **FTR:** First-Time-Right (Metric tracking defect-free installations).
- **UAT:** User Acceptance Testing.
- **SAT:** Site Acceptance Testing.
- **CAB:** Change Advisory Board.

Assumptions:

- Quality controls apply equally to both internal VI personnel and authorized sub-contracted ecosystem partners.
- The framework strictly aligns with ITIL v4 Release Management best practices.

Applicability:

- **Target Audience:** Bid Writers answering "Quality Assurance" RFP sections, Project Managers, and Solution Architects.
- **Scope:** Managed Network Services provisioning, Security baseline deployments, and complex SIP/Voice integrated porting.

3. Core Quality Objectives

VI Business tracks the overall success of the QMF against four immutable, Board-level KPIs:

1. **First-Time-Right (FTR) Delivery:** Target > 98.5%. A deployment is only considered FTR if the physical installation and logical configuration succeed on the first scheduled dispatch without requiring rework.
2. **Zero Critical Defect Leakage:** No P1 (Critical) network incidents originating from a newly deployed site within the initial 7-day Hypercare window.
3. **Process Adherence:** 100% compliance with mandatory staging and burn-in testing prior to hardware dispatch.

4. **Customer Satisfaction (CSAT):** Achieving a minimum 4.5/5 rating on the automated post-deployment project closure survey.

4. The Quality Review Checkpoints

Quality injection begins at the pre-sales phase and continues through to billing handover.

- **Checkpoint 1: Solution Engineering Validation (Pre-Sales):** The central Deal Desk categorically rejects any structural Bill of Materials (BoM) that lacks a countersignature from a certified Tier-3 Solution Architect verifying architectural feasibility against core capacity.
- **Checkpoint 2: Pre-Staging Burn-In (Logistics):** Before Customer Premises Equipment (CPE) leaves the Bhiwandi warehouse, it undergoes a 12-hour electrical power-on burn-in test, and factory default passwords are wiped.
- **Checkpoint 3: The Go/No-Go Bridge (Deployment):** Immediately prior to bringing a site live on the network, the Project Manager polls the Field Engineer, the NOC L2 Engineer, and the Customer IT Lead. If any parameter is red, deployment is unconditionally aborted.

5. Testing Quality Gates (UAT & SAT)

- **Site Acceptance Testing (SAT):** Conducted by the localized Field Engineer. Validates physical metrics: neat rack cabling, properly labeled power inputs, and successful ICMP ping reachability to the central Orchestrator.
- **User Acceptance Testing (UAT):** Conducted jointly by the VI NOC and the Customer IT team. Validates logical metrics: specific application throughput tests (e.g., ensuring SAP traffic correctly routes via the premium MPLS underlay rather than the standard internet backup).
- **Sign-Off:** Only after secure, digitally signed completion of the SAT/UAT checklist via the Field Service Management app does the service formally shift to a "Billing Active" state.

6. Document Control and Configuration Management

- **Version Superiority:** The customer's tenant configuration logic residing within the central VI Orchestrator (e.g., vManage) is deemed the absolute "Single Source of Truth."
- **Prohibition on Local CLI:** Local Field Engineers are strictly banned from making manual Command Line Interface (CLI) changes to the CPE. 100% of configuration changes must be pushed centrally from the Orchestrator by the NOC to prevent configuration drift.
- **Configuration Backups:** Automated backups of all customer tenant databases are executed nightly and stored in immutable, air-gapped AWS S3 buckets in the Mumbai `ap-south-1` region.

7. Audit and Compliance Reviews

Quality assurance requires policing.

- **Internal CAB Audits:** A random 10% sampling of all completed "Standard Change" tickets is audited monthly by the PMO to ensure engineers are not bypassing structural approvals.
- **External ISO Surveillance:** The entire delivery and operations lifecycle is subjected to annual third-party external audits by recognized certifiers (e.g., BSI) to maintain the corporate ISO 9001:2015 (Quality) and ISO 20000-1 (Service Management) credentials.

8. The Continuous Improvement (CSI) Loop

- **Post-Incident Reviews (PIR):** Any deployment resulting in a failure (FTR breach) automatically spawns a Problem Management ticket.

- **Root Cause Analysis:** L3 SMEs must identify whether the failure was logistical, human-error, or a software bug.
- **Process Remediation:** If human error is identified, the standard operating procedure (SOP) is rewritten, and the relevant engineering pod undergoes mandatory re-training within 14 days.

9. Constraints, Exclusions, and Dependency Notes

- **Customer-Supplied Hardware:** If a customer insists on retaining legacy, depreciated edge hardware (which VI Business agrees to manage strictly under a "Best Effort" waiver), VI explicitly disclaims adherence to the FTR metrics for that specific site rollout due to unknown underlying hardware degradation.

10. Proposal-Use Guidance

- **RFP Quality Sections:** Exploit Section 4 (The Checkpoints) and Section 6 (Prohibition on Local CLI) heavily. These concrete technical policies score far higher with technical evaluators than generic "we are committed to quality" statements.

11. Governance and Ownership

- **Owner:** National Head of Quality Assurance & PMO
- **Reviewers:** IT Service Delivery, Deals Desk.
- **Update Cadence:** Annual alignment with ISO 9001 standards.